	Criteria	Levels of Performance				Expected Components	SLOs
		Expert (4)	Proficient (3)	Developing (2)	Novice (1)		
General	**Overall Delivery and Presentation**	Professional, well-organized, succinct, and detailed delivery. The submission meets all the requirements outlined, visually appealing, clear language, appropriate referencing and length.	Organized delivery but needs impovement in some areas. The submission is somewhat visually appealing with some inconsistencies, clear language with minor errors, appropriate referencing and length.	Slightly disorganized delivery. The submission is visually slightly confusing with unclear language and some errors, provides some unnecessary/insufficient information, limited referencing, or is over-length.	Disorganized delivery. The submission is visually confusing/limited/poor with fluffy language and numerous errors. Not enough or overwhelming content. No references.	1. Document structure: consistent layout/font/sectioning 2. High-quality graphics 3. Logical layout of tables and lists 4. Grammatically and typographically correct language 5. Accurate referencing if required	SLO4
General	**Personal Profile**	The personal profile is exceptionally well-written, is engaging, and covers all the required content in a professional manner: Exceptionally clear and concise introduction, engineering skills highlighted, high-quality photo, and insightful motivation and goals.	The personal profile is complete and well-written and covers all required contents: Clear introduction, engineering skills highlighted, good quality photo, and clear motivation and goals.	The personal profile is presented, but there are areas that could be improved in regard to the required content: Introduction needs clarity, basic engineering skills highlighted, photo may be of lower quality, and motivation and goals are somewhat unclear.	The personal profile is incomplete or missing key information of the required contents: Introduction is missing or very unclear, engineering skills not highlighted, no or low-quality photo, and lacks motivation and goals.	1. Personal introduction and background 2. High-quality photo 3. Motivation and learning goals for 41070 EMS	SLO4
General	**Team Role and Contribution**	Provides a thorough and insightful description on the student's role in the project team, showing a deep understanding of their contributions and impact. Clearly explains their tasks, achievements, and leadership with evidence from communication/collaboration tools.	Presents a well-developed description on the student's role in the project team, describing contributions and responsibilities effectively. Details tasks, achievements, and collaboration with evidence from communication/collaboration tools.	Offers a basic description on the student's role, with limited insight into contributions and responsibilities. Describes tasks and interactions at a surface level with minimal evidence from communication/collaboration tools.	Provides an inadequate description on the student's role, lacking clarity and coherence. Fails to articulate contributions and responsibilities effectively without any evidence from communication/collaboration tools.	1. Team role and workload distribution 2. Contribution to the project 3. Discipline lead specific contribution 4. Evidence from communication/collaboration tools	SLO4
General	**Subsystem Overview and Specifications**	A comprehensive and detailed overview of the subcircuit and subroutine is presented, covering all key components and functionalities (system architecture, hardware, software). A clear and comprehensive set of requirements in the form of tables/lists is presented with detailed and precise block diagrams.	An overview of the subcircuit and subroutine is presented, addressing most key components and functionalities (system architecture, hardware, software). A sufficient but partially incomplete set of requirements in the form of tables/lists is presented. Block diagrams are presented but lack details and/or there is room for improvement.	A basic overview of the subcircuit and subroutine is presented, covering essential components and functionalities, but lacking detail of the system architecture, hardware, and/or software. A limited set of requirements is presented and block diagrams are underdeveloped but mostly complete.	A superficial or incomplete overview of the subcircuit and subroutine is presented with missing key components and/or functionalities. There is a significant lack of understanding and/or inaccuracies/omissions in the system architecture, hardware, and/or software. Requirements and/or block diagrams are incomplete or missing entirely.	1. Subcircuit and subroutine introduction 2. Functional and hardware requirements list 3. Block diagrams	SLO1
ECAD	**Subcircuit Schematic Design**	Clear, detailed, complete, and correct subschematic with annotations highlighting design choices supported by clear explanations, calculations, simulations of the subcircuit. All components are annotated with values and part numbers.	Complete and correct subschematic with some missing annotations or explanations, calculations, simulations of subcircuit. Most components are annotated with values and part numbers.	Complete and mostly correct subschematic but annotations / explanations are mostly missing and calculations, simulations of subcircuit are preliminary. Only some components are annotated with values and part numbers.	Incomplete or severly incorrect subschematic without sufficient explanations, calculations, simulations of subcircuit. Most components are not annotated, have missing values or part numbers.	1. Schematic of subcircuit with annotations 2. Explanations, calculations, simulations of subcircuit	SLO2 SLO3
ECAD	**Subcircuit PCB Design**	The PCB layout of the subcircuit demonstrates exceptional understanding of electronic principles with optimal component placement and routing for efficient signal flow and minimal noise. Components are placed logically and routed efficiently. Design rules are well explained, chosen logically and adhered to. All components and footprints have been carefully selected and justified.	The PCB layout of the subcircuit shows a solid understanding of electronic principles with appropriate component placement and routing for efficient signal flow. Components are mostly placed logically and routed effectively. Design rules are explained and adhered to but may be too critical or loose. Most components and footprints have been carefully selected and justified.	The PCB layout of the subcircuit displays some weaknesses in understanding electronic principles with suboptimal component placement and routing leading to potential signal integrity issues. Component placement shows defficiences or is cluttered and routing can be significantly improved. Design rules are only loosley explained and sometimes not adhered to. Only some components and footprints have been carefully selected and justified.	The PCB layout of the subcircuit lacks understanding of electronic principles with poor component placement and routing resulting in significant signal integrity issues. Components and/or trace placement is erroneous and compromises performance. Design rules are not established or severly incomplete. Most components and/or footprints have not been carefully selected and justified.	1. PCB layout of subcircuit 2. Component placement, routing, vias 3. Design rules 4. Component and footprint selection	SLO2 SLO3
ECAD	**Subcircuit Testing and Validation**	Rigorous testing and validation procedures are conducted on the assembled subcircuit, including functional testing, electrical testing, and inspection for defects. A comprehensive test report is included that demonstrates how the subcircuit was tested and validated along with its specifications. Electrical faults are properly documented, reflected on, fixed and / or noted for a revision.	Testing and validation procedures are performed on the assembled subcircuit, covering essential parameters such as functionality, electrical continuity, and visual inspection for defects. A test report is included that demonstrates how the subcircuit was tested and validated along with its specifications. Most electrical faults are documented, reflected on, fixed and / or noted for a revision.	Limited testing and validation are conducted on the subcircuit, focusing mainly on functionality and electrical continuity. Inspection for defects may be overlooked. A basic test report is included that demonstrates how the subcircuit was tested and validated along with its specifications. Some electrical faults are documented, reflected on, fixed and / or noted for a revision.	Testing and validation are minimal or absent, leaving the reliability and functionality of the subcircuit uncertain. A test report is severly incomplete and/or validation with specifications is missing. Electrical faults are not properly documented or addressed.	1. PCB subsystem testing & validation 2. Test report with all specifications / faults recorded	SLO1 SLO3 SLO4
Software	**Embedded Software Subroutine**	Thorough and well-organized documentation for the embedded software subroutine is provided, including detailed explanations of code architecture, modules, and functions. The user guide is easy to follow and the program flow chart is logical.	Clear and structured documentation for the embedded software subroutine is provided, including explanations of key code architecture, modules, and functions. The user guide is mostly easy to follow and the program flow chart is mostly logical with minor inconsistencies.	Basic documentation for the embedded software subroutine is provided, outlining code structure, modules, and functions. The user guide and flow charts are, in parts, difficult to follow or lack logic with several inconsistencies.	Inadequate documentation for the embedded software subroutine is provided, lacking structure and detail. The user guide is mostly difficult to follow and/or the program flow chart is confusing or missing entirely.	1. Quick user guide 2. Subroutine flow charts	SLO2 SLO3
Software	**Embedded Software Subroutine Testing and Validation**	Comprehensive testing and validation procedures are executed for the embedded software subroutine, covering all functionalities and edge cases. Rigorous testing is performed using various methods and tools, ensuring high code coverage and identifying potential issues. The subroutine is validated against requirements and specifications, ensuring compliance and robustness.	Thorough testing and validation of the embedded software subroutine is performed, addressing most functionalities and common scenarios. Testing using appropriate methods and tools is performed, achieving satisfactory code coverage and detecting most issues. The subroutine is validated against requirements, ensuring functionality and reliability.	Basic testing and validation of the embedded software subroutine is performed, covering essential functionalities and limited scenarios. Tests are executed using available methods and tools, achieving minimal code coverage and identifying some issues. The subroutine is validated against basic requirements, ensuring basic functionality.	Inadequate testing and validation of the embedded software subroutine is performed, missing critical functionalities and major scenarios. Testing fails to achieve sufficient code coverage and overlooks many issues. Subroutine is not validated against requirements, leading to potential functionality and reliability issues.	1. Embedded software subroutine testing & validation 2. Test report with all specifications recorded	SLO1 SLO3 SLO4
Reflection	**Evaluation, feedback and reflection**	Comprehensive reflection including detailed design review feedback and self evaluation with insightful learning outcomes. Demonstration of comprehensive understanding of feedback received and implementation of specific actions taken to address areas for improvement. Offers critical analysis and self-awareness, highlighting strengths, challenges, and lessons learned from the experience.	Good reflection including design review feedback and self evaluation with good learning outcomes. Demonstration of good understanding of the feedback received and implementation of some actions taken to address areas for improvement. Provides some analysis of their strengths and areas for improvement, challenges, and lessons learned from the experience.	Adequate reflection with some design review feedback and basic learning outcomes. Demonstration of an acceptable level of understanding of the feedback received with actions taken to address areas for improvement. Shows minimal analysis of personal performance, indicating a need for further self-reflection.	Limited reflection, minimal design review feedback and unclear learning outcomes. Insufficient demonstration of understanding of the feedback received and missing evidence of actions taken to address areas for improvement. Offers no or minimal analysis of personal performance.	1. Group evaluation 2. Peer feedback 3. Weekly reflection journal	SLO5